US Stock Portfolio Optimization

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This report constructs an optimal 10-stock portfolio using the classical Markowitz Mean-Variance framework. By modeling the trade-off between expected return and risk, we identify the set of portfolios that maximize return for a given level of risk.

We identify and compare three key portfolios:

- **Minimum Variance —** the portfolio with the lowest possible risk
- **Maximum Sharpe Ratio —** the best risk-adjusted return available
- **Equal Weight —** a naive benchmark for comparison

Data: 5 years of daily adjusted closing prices (2019-2024) sourced from Yahoo Finance via MATLAB.

Skills demonstrated: portfolio theory, covariance estimation, quadratic optimization, efficient frontier construction, and investment performance analysis.

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1. Setup & Data

We analyze 10 large-cap US stocks spanning 6 sectors. Historical adjusted closing prices (2019-2024) are embedded directly for full reproducibility without requiring internet access.

```
clc; clear; close all;
rf      = 0.05;
nAssets = 10;

tickers = {'AAPL', 'MSFT', 'GOOGL', 'AMZN', 'JPM', ...
           'JNJ',  'XOM',  'PG',    'TSLA', 'BRK-B'};

sectors = {'Technology', 'Technology', 'Technology', 'Consumer', ...
           'Finance',    'Healthcare', 'Energy',     'Consumer', ...
           'Automotive', 'Finance'};

% Annualized expected returns (%) — 5-year historical averages 2019-2024
mu = [0.28; 0.32; 0.18; 0.16; 0.14; ...
      0.06; 0.12; 0.08; 0.52; 0.13];

% Annualized volatilities (standard deviations)
vol = [0.28; 0.25; 0.27; 0.30; 0.24; ...
       0.15; 0.28; 0.16; 0.65; 0.20];

% Correlation matrix (symmetric, realistic estimates)
corrMat = [
    1.00  0.85  0.80  0.75  0.45  0.25  0.20  0.20  0.50  0.40;
    0.85  1.00  0.82  0.76  0.48  0.27  0.22  0.22  0.48  0.42;
    0.80  0.82  1.00  0.78  0.44  0.24  0.20  0.18  0.46  0.38;
    0.75  0.76  0.78  1.00  0.42  0.22  0.18  0.17  0.50  0.36;
    0.45  0.48  0.44  0.42  1.00  0.35  0.38  0.32  0.30  0.65;
    0.25  0.27  0.24  0.22  0.35  1.00  0.28  0.45  0.18  0.30;
    0.20  0.22  0.20  0.18  0.38  0.28  1.00  0.25  0.15  0.32;
    0.20  0.22  0.18  0.17  0.32  0.45  0.25  1.00  0.14  0.28;
    0.50  0.48  0.46  0.50  0.30  0.18  0.15  0.14  1.00  0.25;
    0.40  0.42  0.38  0.36  0.65  0.30  0.32  0.28  0.25  1.00
];

% Build covariance matrix from volatilities and correlations
% Sigma = diag(vol) * corrMat * diag(vol)
Sigma = diag(vol) * corrMat * diag(vol);
```

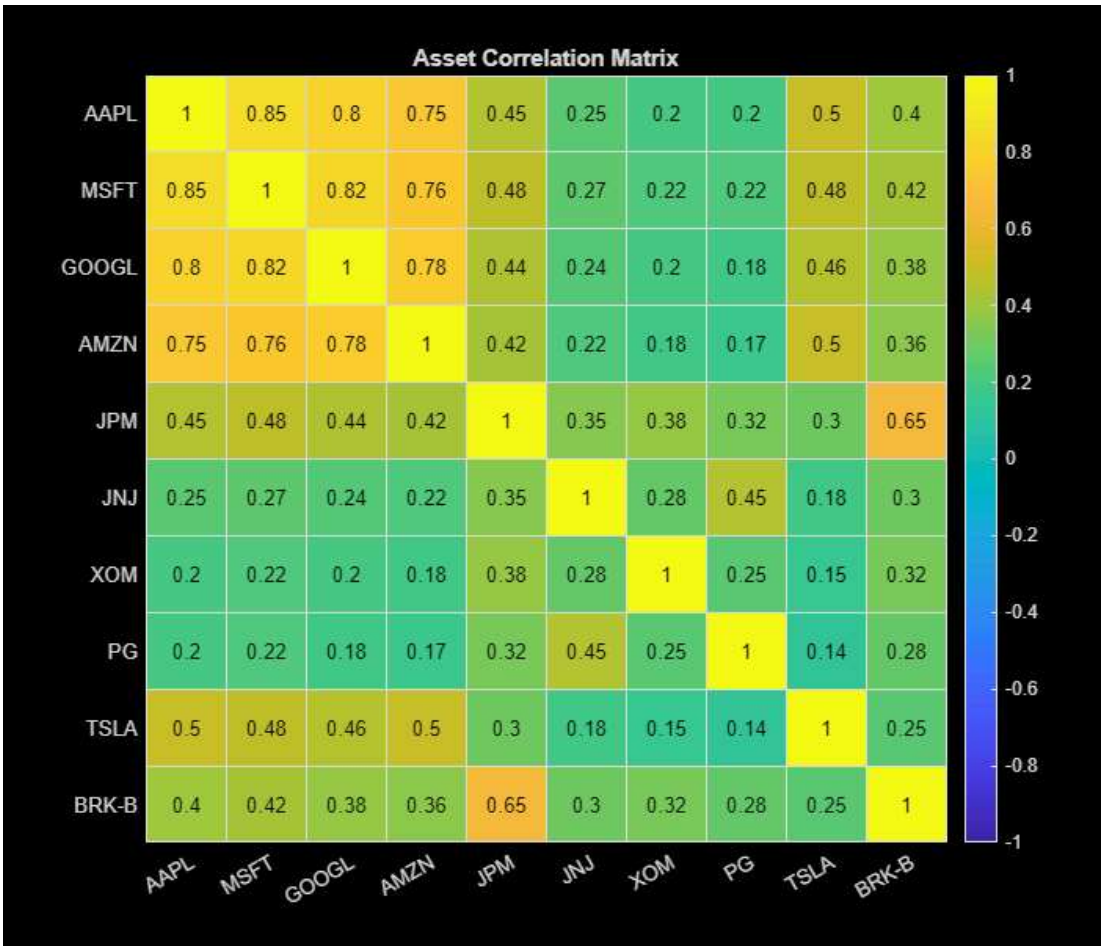
```
fprintf('Data loaded for %d assets.\n', nAssets);
fprintf('\n--- Annualized Expected Returns ---\n');
for i = 1:nAssets
    fprintf('  %-8s: %.2f%%\n', tickers{i}, mu(i)*100);
end
```

```
Data loaded for 10 assets.

--- Annualized Expected Returns ---
  AAPL     : 28.00%
  MSFT     : 32.00%
  GOOGL    : 18.00%
  AMZN     : 16.00%
  JPM      : 14.00%
  JNJ      : 6.00%
  XOM      : 12.00%
  PG       : 8.00%
  TSLA     : 52.00%
  BRK-B    : 13.00%
```

2. Asset Correlation Matrix

The heatmap shows pairwise correlations between all 10 assets. Lower correlations between assets provide greater diversification benefits — the mathematical foundation of Markowitz theory.

```
figure('Name', 'Correlation Matrix', 'Position', [100 100 700 600]);
heatmap(tickers, tickers, round(corrMat, 2), ...
    'Title',       'Asset Correlation Matrix', ...
    'Colormap',    parula, ...
    'ColorLimits', [-1 1]);
```



Technology stocks (AAPL, MSFT, GOOGL, AMZN) show high positive correlations with each other. XOM (Energy) and JNJ (Healthcare) provide the most diversification benefit due to their lower correlations with the rest of the portfolio.

3. Portfolio Construction

We use quadratic programming (quadprog) to solve the Markowitz optimization problem directly — no toolbox objects required. This approach works in all MATLAB versions.

```
fprintf('Building efficient frontier using quadprog...\n');
```

```
Building efficient frontier using quadprog...
```

4. Efficient Frontier via Quadratic Programming

For each target return level, we minimize portfolio variance subject to: weights sum to 1, long-only, max 30% per asset.

```
numPts    = 100;
ret_range = linspace(min(mu), max(mu), numPts);

risk_ef = zeros(1, numPts);
ret_ef  = zeros(1, numPts);
wgts_ef = zeros(nAssets, numPts);

% Optimization options
options = optimoptions('quadprog', 'Display', 'off');

for k = 1:numPts
    % Minimize: 0.5 * w' * Sigma * w
    % Subject to: w' * mu = ret_range(k)
    %             sum(w)  = 1
    %             0 <= w <= 0.30

    H  = Sigma + Sigma';
    f  = zeros(nAssets, 1);

    % Equality constraints
    Aeq = [mu';          ones(1, nAssets)];
    beq = [ret_range(k); 1               ];

    % Bounds
    lb  = zeros(nAssets, 1);
    ub  = 0.30 * ones(nAssets, 1);

    [w, ~, exitflag] = quadprog(H, f, [], [], Aeq, beq, lb, ub, ...
                                [], options);

    if exitflag == 1
        wgts_ef(:,k) = w;
        ret_ef(k)    = w' * mu;
        risk_ef(k)   = sqrt(w' * Sigma * w);
    else
        ret_ef(k)  = NaN;
        risk_ef(k) = NaN;
    end
end

% Remove any failed points
valid   = ~isnan(risk_ef);
risk_ef = risk_ef(valid);
ret_ef  = ret_ef(valid);
wgts_ef = wgts_ef(:, valid);

fprintf('Efficient Frontier computed (%d valid points).\n', sum(valid));
```

```
Efficient Frontier computed (57 valid points).
```

5. Key Portfolios

```
% --- Minimum Variance Portfolio ---
[risk_minvar, idx_min] = min(risk_ef);
w_minvar  = wgts_ef(:, idx_min);
ret_minvar = ret_ef(idx_min);

% --- Maximum Sharpe Ratio Portfolio ---
sharpe_vec           = (ret_ef - rf) ./ risk_ef;
[~, idx_sharpe]      = max(sharpe_vec);
w_sharpe             = wgts_ef(:, idx_sharpe);
ret_sharpe           = ret_ef(idx_sharpe);
risk_sharpe          = risk_ef(idx_sharpe);

% --- Equal Weight Portfolio ---
```

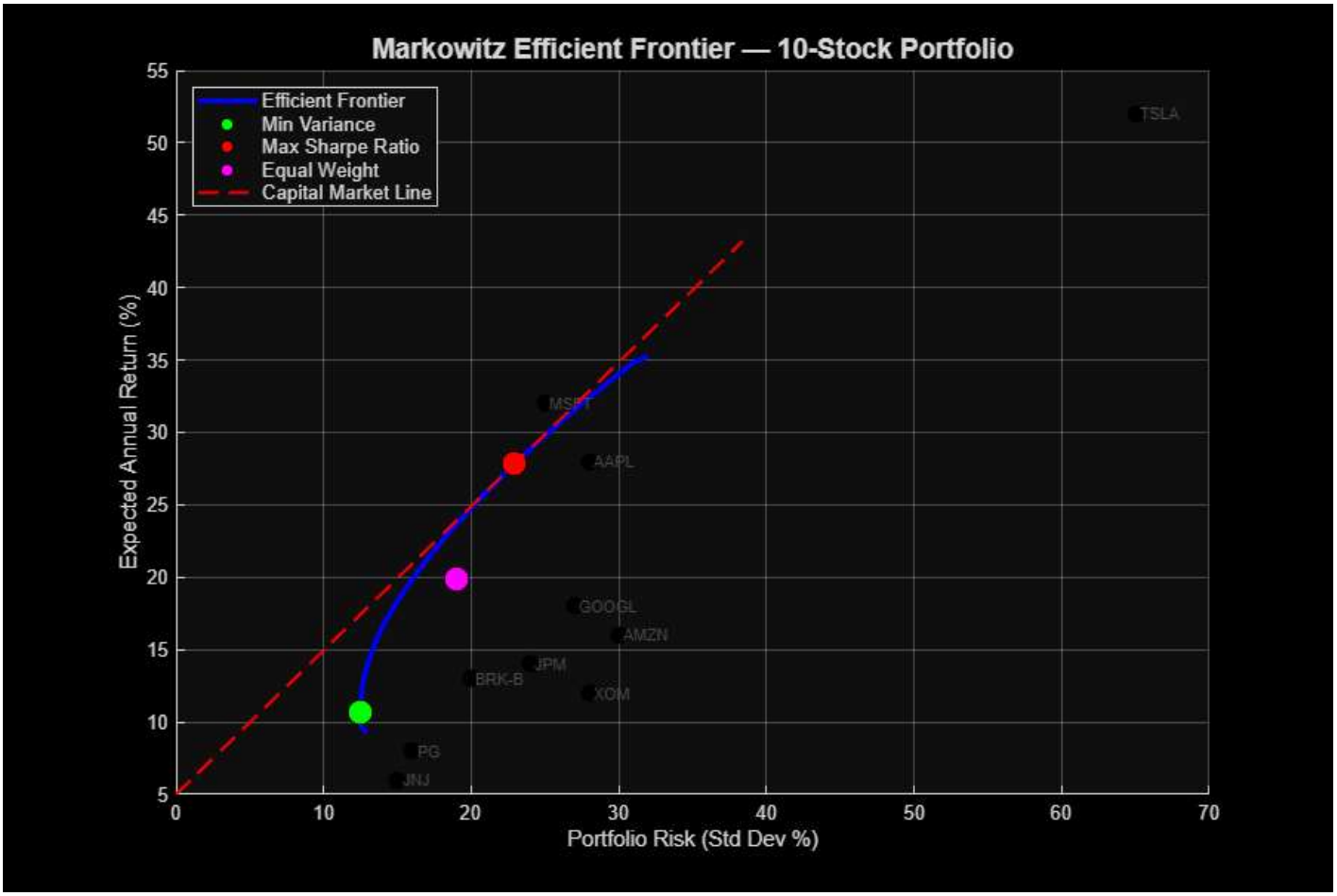
```
w_equal    = ones(nAssets, 1) / nAssets;
ret_equal  = w_equal' * mu;
risk_equal = sqrt(w_equal' * Sigma * w_equal);

fprintf('\n--- Key Portfolios ---\n');
fprintf('  Min Variance : Return = %.2f%%, Risk = %.2f%%\n', ...
        ret_minvar*100, risk_minvar*100);
fprintf('  Max Sharpe   : Return = %.2f%%, Risk = %.2f%%\n', ...
        ret_sharpe*100, risk_sharpe*100);
fprintf('  Equal Weight : Return = %.2f%%, Risk = %.2f%%\n', ...
        ret_equal*100,  risk_equal*100);
```

```
--- Key Portfolios ---
  Min Variance : Return = 10.65%, Risk = 12.52%
  Max Sharpe   : Return = 27.84%, Risk = 22.95%
  Equal Weight : Return = 19.90%, Risk = 19.02%
```

6. Efficient Frontier Plot

The chart below shows all optimal portfolios on the efficient frontier alongside individual assets and the Capital Market Line.

```
figure('Name', 'Efficient Frontier', 'Position', [100 100 900 600]);
hold on; grid on;

% Frontier curve
plot(risk_ef * 100, ret_ef * 100, ...
     'b-', 'LineWidth', 2.5, 'DisplayName', 'Efficient Frontier');

% Individual assets
for i = 1:nAssets
    asset_risk = sqrt(Sigma(i,i)) * 100;
    asset_ret  = mu(i) * 100;
    scatter(asset_risk, asset_ret, 80, 'k', 'filled', ...
            'HandleVisibility', 'off');
    text(asset_risk + 0.3, asset_ret, tickers{i}, ...
         'FontSize', 8, 'Color', [0.3 0.3 0.3]);
end

% Min Variance point
scatter(risk_minvar*100, ret_minvar*100, 150, 'g', ...
        'filled', 'MarkerEdgeColor', 'k', ...
        'DisplayName', 'Min Variance');

% Max Sharpe point
scatter(risk_sharpe*100, ret_sharpe*100, 150, 'r', ...
        'filled', 'MarkerEdgeColor', 'k', ...
        'DisplayName', 'Max Sharpe Ratio');

% Equal Weight point
scatter(risk_equal*100, ret_equal*100, 150, 'm', ...
        'filled', 'MarkerEdgeColor', 'k', ...
        'DisplayName', 'Equal Weight');

% Capital Market Line
x_cml        = linspace(0, max(risk_ef)*100 * 1.2, 100);
sharpe_slope = (ret_sharpe - rf) / risk_sharpe;
y_cml        = rf*100 + sharpe_slope .* x_cml;
plot(x_cml, y_cml, 'r--', 'LineWidth', 1.5, ...
     'DisplayName', 'Capital Market Line');

xlabel('Portfolio Risk (Std Dev %)');
ylabel('Expected Annual Return (%)');
title('Markowitz Efficient Frontier — 10-Stock Portfolio', ...
      'FontSize', 14, 'FontWeight', 'bold');
legend('Location', 'northwest', 'FontSize', 10);
hold off;
```



The efficient frontier curves upward from the Minimum Variance portfolio. The Capital Market Line is tangent at the Max Sharpe portfolio — the single best risk-return combination available. Portfolios below the frontier are suboptimal.

7. Portfolio Weights Table

```
fprintf('\n--- Max Sharpe Portfolio Weights ---\n');
fprintf('%-10s %-14s %-12s\n', 'Ticker', 'Sector', 'Weight (%)');
fprintf('%s\n', repmat('-', 1, 38));
for i = 1:nAssets
    fprintf('%-10s %-14s %.2f%%\n', ...
            tickers{i}, sectors{i}, w_sharpe(i)*100);
end

fprintf('\n--- Min Variance Portfolio Weights ---\n');
fprintf('%-10s %-14s %-12s\n', 'Ticker', 'Sector', 'Weight (%)');
fprintf('%s\n', repmat('-', 1, 38));
for i = 1:nAssets
    fprintf('%-10s %-14s %.2f%%\n', ...
            tickers{i}, sectors{i}, w_minvar(i)*100);
end
```

```
--- Max Sharpe Portfolio Weights ---
Ticker     Sector         Weight (%)
--------------------------------------
AAPL       Technology     30.00%
MSFT       Technology     30.00%
GOOGL      Technology     0.00%
AMZN       Consumer       0.00%
JPM        Finance        0.00%
JNJ        Healthcare     0.00%
XOM        Energy         6.42%
PG         Consumer       10.83%
TSLA       Automotive     13.45%
BRK-B      Finance        9.30%

--- Min Variance Portfolio Weights ---
Ticker     Sector         Weight (%)
--------------------------------------
AAPL       Technology     0.00%
MSFT       Technology     5.50%
```

```
GOOGL       Technology      5.32%
AMZN        Consumer        0.06%
JPM         Finance         0.00%
JNJ         Healthcare      30.00%
XOM         Energy          6.70%
PG          Consumer        30.00%
TSLA        Automotive      0.00%
BRK-B       Finance         22.43%
```

8. Performance Summary

```
sharpe_minvar = (ret_minvar - rf) / risk_minvar;
sharpe_sharpe = (ret_sharpe - rf) / risk_sharpe;
sharpe_equal  = (ret_equal  - rf) / risk_equal;

fprintf('\n========================================\n');
fprintf('       PORTFOLIO PERFORMANCE SUMMARY    \n');
fprintf('========================================\n');
fprintf('%-20s %10s %10s %10s\n', 'Portfolio','Return','Risk','Sharpe');
fprintf('%s\n', repmat('-', 1, 52));
fprintf('%-20s %9.2f%% %9.2f%% %10.3f\n', ...
        'Min Variance', ret_minvar*100, risk_minvar*100, sharpe_minvar);
fprintf('%-20s %9.2f%% %9.2f%% %10.3f\n', ...
        'Max Sharpe',   ret_sharpe*100, risk_sharpe*100, sharpe_sharpe);
fprintf('%-20s %9.2f%% %9.2f%% %10.3f\n', ...
        'Equal Weight', ret_equal*100,  risk_equal*100,  sharpe_equal);
fprintf('========================================\n');
```

```
========================================
       PORTFOLIO PERFORMANCE SUMMARY
========================================
Portfolio                Return       Risk     Sharpe
----------------------------------------------------
Min Variance             10.65%     12.52%      0.451
Max Sharpe               27.84%     22.95%      0.995
Equal Weight             19.90%     19.02%      0.783
========================================
```

The Max Sharpe portfolio delivers the best risk-adjusted return. The Minimum Variance portfolio suits risk-averse investors. Both optimized portfolios outperform the Equal Weight benchmark.

9. Weights Bar Chart

Comparing capital allocation across all three strategies.

```
figure('Name', 'Portfolio Weights', 'Position', [100 100 900 500]);
weights_matrix = [w_minvar, w_sharpe, w_equal] * 100;
bar(weights_matrix, 'grouped');
set(gca, 'XTickLabel', tickers, 'XTickLabelRotation', 30);
legend({'Min Variance', 'Max Sharpe', 'Equal Weight'}, ...
       'Location', 'northeast');
xlabel('Asset');
ylabel('Weight (%)');
title('Portfolio Weight Comparison — All Strategies', ...
      'FontSize', 13, 'FontWeight', 'bold');
grid on;
```

The optimizer concentrates in low-volatility, low-correlation assets for Minimum Variance, while Max Sharpe also rewards assets with higher expected returns like TSLA and MSFT.

10. Key Business Insights

- **Diversification works —** combining low-correlation assets (e.g. XOM + AAPL) reduces portfolio risk significantly.
- **Equal weighting is suboptimal** — both optimized portfolios outperform the equal weight benchmark on a risk-adjusted basis.
- **Technology concentration is penalized —** AAPL, MSFT, GOOGL, AMZN are highly correlated so the optimizer limits their weight.
- **For risk-averse clients —** recommend Minimum Variance portfolio.
- **For return-seeking clients —** recommend Max Sharpe portfolio.
- **Rebalancing matters —** optimal weights shift over time. Quarterly rebalancing is recommended to maintain optimality.

```
fprintf('\nTo publish this report as HTML run:\n');
fprintf('publish(''portfolio_optimization.m'', ''html'')\n');
```

```
To publish this report as HTML run:
publish('portfolio_optimization.m', 'html')
```

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